

Gen Digital

Norton Mobile Engineering

AI-First Intern – Take-Home Assignment

Duration: 7 days | Estimated effort: 8–12 hours
Internship: 3-month AI-First Mobile Engineering

Gen Digital Inc. – Norton Brand Products
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1. Overview

Welcome to the Gen Digital - Norton Mobile Engineering AI-First Intern take-home assignment. This exercise is designed to evaluate your ability to use AI tools as a core part of your development workflow — not just as an add-on. During the 3-month internship, you will work on Norton 360 mobile apps across iOS and Android, and we expect you to leverage AI coding assistants daily for code generation, debugging, testing, and code review.

This assignment has three goals:

- Assess your proficiency with AI-assisted development (Claude, GitHub Copilot, Cursor, or similar)
- Evaluate your mobile development fundamentals (Swift/Kotlin/Compose Multiplatform/Flutter/React Native)
- Gauge your understanding of cybersecurity product concepts relevant to Norton 360

2. Getting Started: Download Norton 360

Before you begin coding, install and explore the Norton 360 app. This will give you hands-on familiarity with the product you will be working on during the internship.

Download on iOS

1. Open the App Store on your iPhone or iPad (requires iOS 17.0 or later).
2. Search for “Norton 360: Security & VPN” (publisher: NortonLifeLock).
3. Tap Get, then Install. Once installed, tap Open.
4. Accept the License Agreement and Global Privacy Statement, then tap Continue.
5. You can explore features using the free trial option — no purchase required.

Download on Android

1. Open the Google Play Store on your Android device (requires Android 10.0 or later).
2. Search for “Norton 360: Virus Scanner & VPN” (publisher: NortonLifeLock).
3. Tap Install. When complete, tap Open.
4. Accept the License Agreement and Global Privacy Statement, then tap Sign In or start a free trial.
5. Grant the requested permissions to enable App Security, Internet Security, and Wi-Fi Security features.

Explore These Features

Spend 20–30 minutes exploring the app. Pay attention to:

- Smart Scan — the primary security scan experience
- Norton Genie — the AI assistant for cybersecurity questions and scam detection
- VPN — one-tap secure connection with bank-grade encryption
- Device Security — OS update alerts and passcode checks
- Internet Security / Safe Web — fraudulent website detection
- Dark Web Monitoring — personal data breach alerts (available on select plans)

Take notes on what you find interesting, any UX patterns you notice, and any areas you think could be improved. You will reference these observations in your assignment.

3. Assignment: Choose One Task

Pick ONE of the two options below. Both are designed to take 8–12 hours and must demonstrate AI-first development practices.

Option A: Build a “Security Health Dashboard” Prototype

Create a mobile app screen (or set of screens) that displays a user’s device security health — inspired by Norton 360’s Smart Scan experience.

Requirements

- Platform: iOS (SwiftUI preferred), Android (Jetpack Compose preferred), or cross-platform (Kotlin Multiplatform / Compose Multiplatform, Flutter, or React Native)
- Display an overall security score (0–100) with a visual indicator (circular progress, gauge, or similar)
- Show at least 4 security check categories (e.g. OS version, app threats, Wi-Fi safety, password strength) with individual status indicators
- Include a “Scan Now” button that triggers a simulated scan with progress animation
- Use mock data — no real backend required; structure your data models as if a real API existed
- Write at least 3 meaningful unit tests covering your data models or view models

Option B: Build a “Scam Message Detector” Prototype

Create a mobile app screen where a user can paste or type a suspicious message (SMS, email snippet, or URL) and get an AI-generated risk assessment — inspired by Norton Genie’s scam detection.

Requirements

- Platform: iOS (SwiftUI preferred), Android (Jetpack Compose preferred), or cross-platform (Kotlin Multiplatform / Compose Multiplatform, Flutter, or React Native)
- Text input field where the user can paste a message or URL
- An “Analyze” button that triggers analysis (you may call a public AI API such as OpenAI or Claude, or use a local heuristic / regex-based approach)
- Display results showing: risk level (Safe / Suspicious / Dangerous), confidence score, and a brief explanation of why the message was flagged
- Show at least 2 example scam messages the user can tap to auto-populate the input field
- Write at least 3 meaningful unit tests covering your analysis logic or view models

4. AI-First Development Requirements

This is the most important section. We are hiring for AI-first engineering skills, and your submission must demonstrate how you use AI tools throughout the development lifecycle.

Mandatory AI Practices

1. Use an AI coding assistant (Claude, GitHub Copilot, Cursor, Windsurf, or similar) throughout your development process.
2. Keep an AI interaction log: In your README, include a dedicated section documenting at least 5 significant prompts you gave the AI and how you used or refined the output. Show your prompt engineering skills.
3. Use AI for test generation: At least one of your unit tests must be AI-generated (with your review and refinement). Note which tests were AI-generated in your code comments.
4. Use AI for code review: Before submitting, ask an AI tool to review your code and document what feedback it gave and what you changed as a result.

What We’re Looking For

- Prompt quality — clear, context-rich prompts that produce useful output on the first or second try
- Critical evaluation — you should not blindly accept AI output; show where you corrected, improved, or rejected AI suggestions
- Iterative refinement — demonstrate follow-up prompts that improved initial results
- Workflow integration — AI should be part of your natural flow, not a one-off gimmick

5. Deliverables & How to Submit

GitHub Repository

1. Create a public GitHub repository named: norton-aifirst-intern-<yourname>
2. Push all source code, tests, and project files to the repository.
3. Include a comprehensive README.md with the following sections:
 - Project overview and which option (A or B) you chose
 - Setup instructions (how to build and run your project)
 - Screenshots of the running app
 - AI Interaction Log (at least 5 prompt/response examples with your commentary)
 - AI Code Review summary (what the AI suggested, what you changed)
 - Reflection: What did you learn? What would you do differently?

Demo Video (5 Minutes)

Record a 5-minute video walkthrough with your camera on (face visible in a corner overlay or side-by-side). The video must cover:

1. App demo (2 min): Walk through your running app, demonstrating all features and interactions.
2. Code walkthrough (1.5 min): Highlight key code decisions, architecture, and how you structured your data models.
3. AI workflow demo (1.5 min): Show how you used AI tools during development — share your screen showing actual AI conversations, explain your prompting strategy, and discuss one example where you refined AI output.

Upload the video to YouTube (unlisted), Google Drive (shared link), or Loom, and include the link in your README.md.

6. How to Share Your Work

6. Email your GitHub repository link and video link to the hiring contact provided in your invitation email.
7. Ensure the GitHub repository is public or the recruiter's GitHub username has been granted access.
8. Ensure the video link is accessible (unlisted YouTube, or Google Drive with "Anyone with the link" sharing). It can be on GitHub too.
9. Double-check that the project builds and runs successfully by cloning it fresh before submitting.

7. Evaluation Criteria

Your submission will be evaluated using the following rubric:

Criteria	Description	Weight
AI Tool Usage	Effective use of AI coding assistants (e.g. Claude, GitHub Copilot, Cursor) for code generation, debugging, and test writing	30%
Code Quality	Clean, well-structured code with proper error handling, naming conventions, and comments	20%
Testing	Meaningful unit tests and/or UI tests with good coverage of edge cases	20%
Product Understanding	Demonstrated understanding of Norton 360 features and cybersecurity domain	15%
Communication	Clarity of README, demo video presentation, and ability to explain AI-assisted workflow	15%

8. Suggested Timeline

You have 7 calendar days from the date you receive this assignment. Here is a suggested approach:

Timeline	Activity
Day 1–2	Download Norton 360, explore features, set up development environment and AI tools
Day 3–4	Choose your task (Option A or B), plan approach, begin AI-assisted development
Day 5–6	Complete implementation, write tests, iterate with AI feedback
Day 7	Record demo video, finalize README, push to GitHub and submit

9. Frequently Asked Questions

Do I need a Norton subscription? No. You can explore the app using the free trial. No purchase is necessary.

Which AI tools can I use? Any AI coding assistant is acceptable: Claude, GitHub Copilot, Cursor, Windsurf, ChatGPT, Gemini, or others. The key is demonstrating effective usage, not which specific tool you choose.

Can I use third-party libraries? Yes. Use whatever packages or libraries you normally would. Just document them in your README and ensure the project builds cleanly.

What if I'm stronger in iOS than Android (or vice versa)? Pick the platform you are most comfortable with. Native iOS (SwiftUI), native Android (Jetpack Compose), or cross-platform (Kotlin Multiplatform / Compose Multiplatform, Flutter, React Native) are all equally welcome. We are assessing your mobile development ability and AI-first workflow, not testing a specific platform.

How polished should the UI be? We value clean, functional UI over flashy design. A well-structured, readable interface with good UX patterns is more impressive than heavy visual polish.

Can I spend more than 12 hours? The assignment is scoped for 8–12 hours. If you find yourself spending significantly more, step back and focus on quality over quantity. We would rather see a well-executed subset than an incomplete attempt at everything.

Good luck! We look forward to seeing your work.

Norton Mobile Engineering Team – Gen Digital Inc.